

WEEK

1

Weekdays until due date: 45

Beginning of week:

- ☐ 1. **Read** and **take notes** on the aim and background information of the BABS1202 Laboratory Project, located in the Class Notebook (Lab 1) on Teams AND the assessment instructions and rubric in the Assessments Hub on Moodle.

Throughout week:

- ☐ 1. **Complete** the lab notebook with the subheadings of date, time, location, collaborators, aim(s) of the lab, protocol, connection to overall project, deviations to protocol, and results and data storage

WEEK

2

Weekdays until due date: 40

Beginning of week:

- ☐ 1. **Create** a laboratory report document and **begin to write** the introduction by using the notes made from the background information and aims - include in-text citations.
2. **Read** the lab project: purification of isolates, located in the Class Notebook (Lab 2) on Teams.

Throughout week:

- ☐ 1. After the lab, **complete** your lab notebook following the Week 1 instructions, using a new tab labelled 'Lab 2'.

End of week:

- ☐ 1. Create a Methods section in your laboratory report document and write:
- a) A subheading for isolating bacteria from soil, and **write a paragraph** describing the isolation of bacteria from soil and the plate dilutions performed in Lab 1.
2. Create a Results section in your laboratory report document and:
- a) **State** the dilution factor plate chosen and CFU.
- b) **Create Table 1** with a legend for the five isolates you chose. Include the Gram staining results and shape, as well as the colony morphology.
- c) **Describe** the table 1 results by referring to the data in a sentence structure.

WEEK

3

Weekdays until due date: 35

Beginning of week:

- ☐ 1. **Continue** writing the introduction and aims for the laboratory report and use relevant **primary/secondary articles**
2. **Read** the lab project: purification of isolates, located in the Class Notebook (Lab 3) on Teams.

Throughout week:

- ☐ 1. After the lab, **complete** your lab notebook following the Week 1 instructions, using a new tab labelled 'Lab 3'.

End of week:

- ☐ 1. Under the Methods section of your laboratory report document :
- a) A subheading for Gram staining and **write a paragraph describing** the Gram staining method from Lab 2
- b) A subheading for Subculture and **write a paragraph describing** the plate streaking performed in Lab 2

WEEK

4

Weekdays until due date: 30

Beginning of week:

- ☐ 1. **Read** the lab project: purification of isolates, located in the Class Notebook (Lab 4) on Teams
2. Continue working on your introduction and aims/ methods and results from Lab 2 if not finished.

Throughout week:

- ☐ 1. After the lab, **complete** your lab notebook following the Week 1 instructions, using a new tab labelled 'Lab 4'.

End of week:

- ☐ 1. Under the Methods section of your laboratory report document:
- a) A subheading for the zone of clearing and **write a paragraph describing** the semi-quantitative analysis of enzyme production performed in lab 4 and iodine flooding performed in lab 3.
- b) A subheading for DNA extraction and PCR setup and **write** a paragraph describing the molecular identification of bacteria by gene sequencing performed in lab 4.

WEEK

5

Weekdays until due date: 25

Beginning of week:

- ☐ 1. **Review and edit** the introduction of your laboratory report.
2. **Read** the lab project: purification of isolates, located in the Class Notebook (Lab 5) on Teams

Throughout week:

- ☐ 1. During the lab, speak to your demonstrator about the report.
2. After the lab, **complete** your lab notebook following the Week 1 instructions, using a new tab labelled 'Lab 5'.

End of week:

- ☐ 1. Under the Methods section of your laboratory report document:
- a) A subheading for electrophoresis and write a paragraph describing the gel electrophoresis of the PCR setup performed in lab 4.
2. Under the Results section of your laboratory report document:
- a) **Add** the photo of the gel and label it Figure 1and include a legend.
- b) **Describe** and refer to Figure 1 gel results in sentence structure – state the size of the amylase band.
- c) **Create Table 2** with the zone of clearing for colony – positive control, negative control and test isolate
- d) **Describe** the table 2 results by referring to the data in a sentence structure.

WEEK

6

Weekdays until due date: 20

Beginning of week:

- ☐
1. Make sure lab notes are up to date – Labs 1-5 have the subheadings of date, time, location, collaborators, aim(s) of the lab, protocol, connection to overall project, deviations to protocol, and results and data storage

Throughout week:

- ☐
1. **Read** the rubric for the introduction, methods and results

2. **Review and edit** the sections you have written so far.

End of week:

- ☐
1. **Fix any errors** or catch up on laboratory report and lab notebook

WEEK

7

Weekdays until due date: 15

Beginning of week:

- ☐
1. **Read** the lab project: purification of isolates, located in the Class Notebook (Lab 6) on Teams.

Throughout week:

- ☐
1. After the lab, **complete** your lab notebook following the Week 1 instructions, using a new tab labelled 'Lab 6'.

2. Begin the discussion by **stating** the aims and the broader significance of the experiment

End of week:

- ☐
1. Under the Methods section of your laboratory report document:

a) A subheading for semi-quantitative measurement of secreted amylase enzyme activity and **write a paragraph describing** the method performed in Lab 6.

b) To the zone of clearing subheading, add the cell-free supernatant drop plate zone of clearing test for enzyme activity method performed in Lab 6.

2. Under the Results section of your laboratory report document:

a) **Add** the standard curve and label it Figure 2 with a legend.

b) **Describe** the table 3 results by referring to the data in a sentence structure

WEEK

8

Weekdays until due date: 10

Beginning of week:

- ☐
1. **Read** the lab project: purification of isolates, located in the Class Notebook (Lab 7) on Teams.

Throughout week:

- ☐
1. After the lab, **complete** your lab notebook following the Week 1 instructions, using a new tab labelled 'Lab 7'.

End of week:

- ☐
1. Under the Results section of your laboratory report document:

a) **Add** the Spectrometric (Abs 595 nm) for starch and Diastix for glucose Table from lab 6, label it as Table 3, and include a legend for the table.

b) **Describe** the table 3 results by referring to the data in a sentence structure.

c) **Add** the graph for starch concentration (ug/mL) vs time (mins) from lab 6, label it as Figure 3 and include a legend.

d) **Describe** the Figure 3 results by referring to the data in a sentence structure

e) **Add** the zone of clearing results to Table 2 for the cell-free extract performed in lab 6 – positive, negative control, test isolate and enzyme mix.

f) **Describe** the zone of clearing for the cell-free extract by referring to Table 2 in sentence structure.

WEEK

9

Weekdays until due date: 5

Beginning of week:

- ☐
1. **Read** the lab project: purification of isolates, located in the Class Notebook (Lab 8) on Teams.

Throughout week:

- ☐
1. After the lab, **complete** your lab notebook following the Week 1 instructions, using a new tab labelled 'Lab 8'.

2. Continue the discussion by summarising the key results from labs 1-8.

End of week:

- ☐
1. Under the methods section of your laboratory report document:

a) A subheading for BLAST search and **write a paragraph describing** the method performed in Lab 8.

2. Under the Results section of your laboratory report document:

a) **State the sentence** provided for BLAST search.

3. For the discussion, **interpret results** by **comparing** the key results from labs 1-8 with relevant scientific literature and theories.

WEEK

10

Due: Friday the 8th August 5pm

Beginning of week:

- ☐
1. Add the title to your report

2. For the discussion, state any sources of error and limitations of the results.

Throughout week:

- ☐
1. Under the conclusion of your laboratory report document, write the conclusion by:

a) Linking the aims of the project to the results

b) Viable applications/significance of the results

c) Future directions

End of week:

- ☐
1. Reference your laboratory report document – make sure to have two scientific articles.

2. Check the rubric for the laboratory report and lab notebook

3. Read the assessment before submitting